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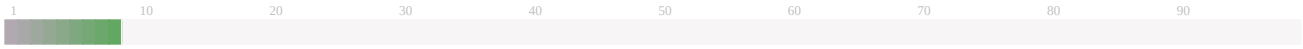
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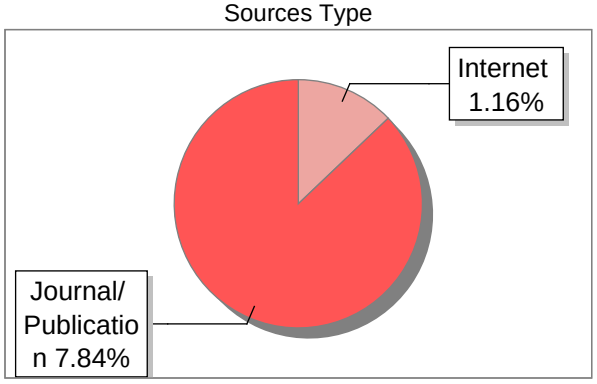
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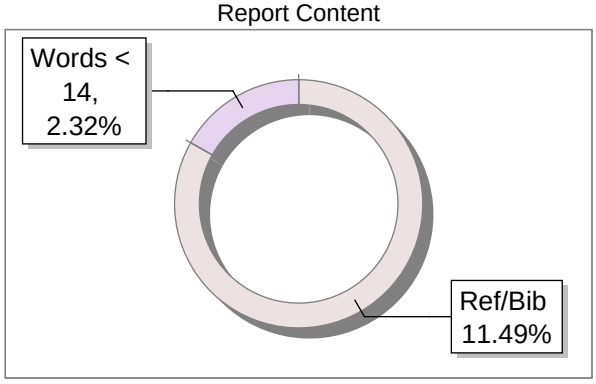
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Explainable AI-Based Clinical Decision Support for Preliminary Dermatological Disease Diagnosis

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Abstract*— *Skin conditions share many common features in appearance, making their diagnosis a challenging task. Deep learning helps to implement skin-image classification tasks, while Explainable AI provides better explanation of predictions [1], [3], [5]. In this paper, we propose an Explainable AI-based Clinical Decision Support System (CDSS) for preliminary skin disease diagnosis. We use dermatological images after preprocessing and use EfficientNet-B0 for Benign/Malignant classification along with confidence measure [4]. For the image localization task,* ***Grad-CAM *is used [5]. Web-based application is considered to provide easy preliminary diagnoses.*

The CDSS is developed to serve as support, but not replacement, of medical specialists, with emphasis on classification accuracy, explainability, and user-friendliness [2].

Key Words*— *Skin Disease Classification, Deep Learning, EfficientNet-B0, CDSS, Explainable AI, Grad-CAM.

I. INTRODUCTION

This paper recognizes the problem in accurate diagnostics of dermatological conditions and the lack of experts' access in underdeveloped regions; hence, AI-powered skin images analytics is proposed [1], [2]. CNNs and other types of deep learning models can extract meaningful visual features, and the publicly accessible dataset HAM10000 can be used to automate the classification process [9]–[11].

Current research studies are dedicated to EfficientNet, transfer learning, attention-based models, transformers, and various explainable AI (XAI) techniques like Grad-CAM and LIME [3]–[7]. However, there are still some doubts

concerning clinical applicability, transparency, and effectiveness on different skin tones [1], [2].

This research proposes an Explainable AI–Based Clinical Decision Support System (CDSS) to diagnose skin conditions. The system combines deep-learning-powered classification with confidence metrics and visual interpretation to help doctors with their diagnoses.

II. RELATED WORK

Recent researches have made use of convolutional neural networks (CNNs) and deep learning in classifying skin diseases automatically [9], [10]. Feature fusion, EfficientNet, attention models, and transformer methods have been used to make further improvements in image-based classification [3], [4], [6], [7]. Grad-CAM and LIME explainable AI approaches have been applied in visualizing the areas that affect model predictions and interpretability [3]-[5]. Furthermore, some researches have considered the applications of clinical decision-making and point of care systems powered by AI for dermatological diagnosis [1], [2], [8]. Overall, the existing literature suggests much potential of AI for skin analyses; however, more works should be done in combining the discussed components.

Table I. Summary of Related Work

Ref.	Approach	Main Focus
[1]	Deep Learning	Skin disease diagnosis across skin tones
[2]	AI Point-of-Care System	Real-time clinical support
[3]	Feature Fusion + XAI	Skin disease classification and interpretability
[4]	Transformer + XAI	Classification and explainability
[5]	CNN + Grad-CAM	Explainable skin disease detection
[6]	EfficientNet + Attention	Multimodal skin disease classification
[8]	AI-based CDSS	Melanoma detection in primary care
[9]	Deep Learning Review	Overview of skin disease recognition

III. RESEARCH GAP

Although there is some literature, a number of problems still arise:

- Lack of XAI in AI: Even though classification models may give precise results, the parts of the images that contribute to the decisions are difficult to interpret [3]-[5].
- Clinical relevance: There are studies that focus on classification rather than integration of predictions with decision support information [1], [2], [8].
- Reporting of prediction confidence: The analysis does not provide prediction confidence information explicitly.
- A need for an integrated framework: A framework combining classification, prediction confidence, visual explanation and decision support is needed.

This research will seek to fill the gaps through the creation of a web-based framework.

V. OBJECTIVES

4A. MAIN OBJECTIVES

To develop an **Explainable Artificial Intelligence-Based Clinical Decision Support System** for the classification of dermatological skin lesions through deep learning, with interpretable predictions and decision support information.

4B. SEPCIFIC OBJECTIVES

- Preprocess and classify skin-lesion images using EfficientNet-B0.
- Solve class imbalance by implementing class-weighted loss function.
- Calculate prediction confidence score for classification results.

- Create visual explanation for predictions with the use of Grad-CAM and LIME.
- Implement model and visual explanations on the web interface.
- Provide preliminary decision support information without replacing professional diagnosis.

V. METHODOLOGY

Workflow of the suggested methodology goes as follows:

Dataset -> Preprocessing -> Lesion-Aware Splitting -> Data Augmentation -> EfficientNet-B0 -> Classification -> Confidence Score -> Grad-CAM/LIME -> Clinical Decision Support -> Web Interface

Processing Steps

1. **Dataset Preparation:** HAM10000 dataset images are divided into benign and malignant classes.

2. **Data Splitting:** Images are split into train, validate and test sets with the help of lesion-aware splitting.

3. **Preprocessing:** Images are resized to 224 × 224 and normalized by the ImageNet distribution.

4. **Data Augmentation:** Train images are flipped horizontally and vertically and rotated.

5. **Model Training:** Pretrained EfficientNet-B0 is trained for binary classification task.

6. **Class Imbalance:** Class-weighted cross entropy loss is used to solve class imbalance problem.

7. **Prediction:** After training the model, the output produced by the model will be a prediction of benign/malignant lesion along with a confidence score.

8. **Explanation:** For explanation of the prediction, Grad-CAM and LIME techniques are used.

9. **Decision Support:** Decision support is provided through the user-friendly web interface.

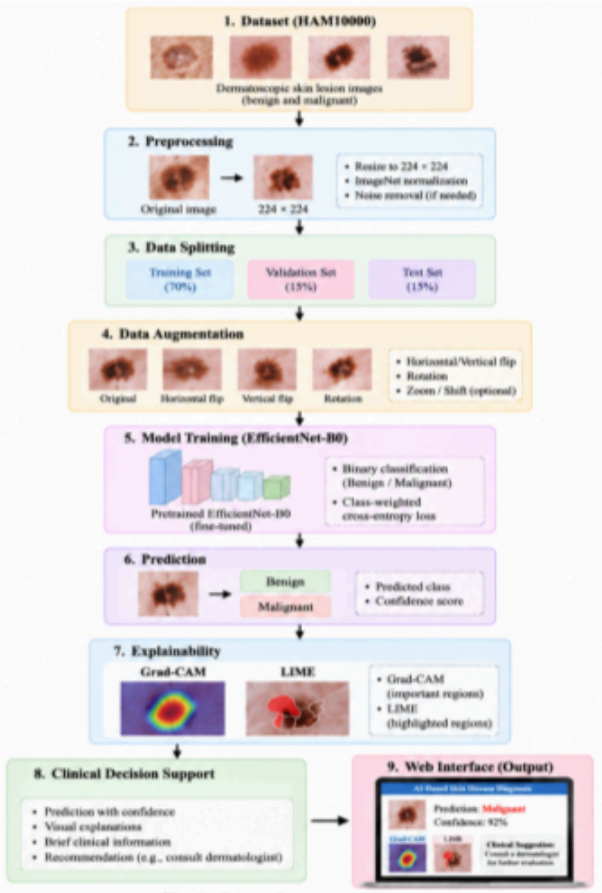


Fig. 1. Proposed system architecture

VI. PROPOSED ALGORITHM

Input: Dermatoscopic skin lesion image

Step 1: Resize the image to 224×224 pixels.

Step 2: Normalize the image using ImageNet statistics.

Step 3: Feed the normalized image to the trained EfficientNet-B0 model.

Step 4: Extract the predicted class – Benign or Malignant.

Step 5: Calculate the confidence of the prediction.

Step 6: Create Grad-CAM and LIME explanations.

Step 7: Display the prediction, confidence and explanation via the web-based interface.

Step 8: Deliver the preliminary decision support data.

Output: Preliminary explainable diagnosis.

VI. DATASET DESCRIPTION

The HAM10000 dataset was used in the dermatological image classification task. The original dataset had been filtered to obtain usable images of **benign and malignant** lesions to finally yield a dataset size of **11,571 images**

Class	Images
Benign	9,415
Malignant	2,156
Total	**11,571**

The dataset was split into **training (6,943), validation (2,314) and test (2,314) sets.** Such a splitting process was done using lesion aware stratification based on lesion IDs to ensure that images from the same lesion do not fall under different subsets. The labels assigned are binary in nature with **0 representing Benign and 1 representing Malignant.**

VII. IMPLEMENTATION DETAILS

The experiment was performed using Python language in Google Colab platform based on PyTorch library with the help of NVIDIA Tesla T4 GPU for the model training.

Parameter	Configuration
Model	Pretrained EfficientNet-B0
Framework	PyTorch
Optimizer	AdamW
Learning Rate	0.0001
Batch Size	16
Epochs	5
Loss Function	Class-weighted Cross-Entropy Loss
Input Size	224 × 224
Classes	Benign, Malignant
Training Platform	Google Colab
GPU	NVIDIA Tesla T4

The final classifier for EfficientNet-B0 model was tuned to give **two outputs** for each image – benign and malignant. The model having the **highest validation F1-score** was saved as best_model.path.

VIII. PERFORMANCE EVALUATION METRICS

The model performance was evaluated using **accuracy, precision, recall, F1-score, and confusion matrix**. Particular attention was given to **malignant-class recall and F1-score** because detecting malignant lesions is important for preliminary screening.

- **Accuracy:** Overall proportion of correctly classified images.
- **Precision:** Proportion of predicted malignant cases that were actually malignant.
- **Recall:** Proportion of actual malignant cases correctly identified.
- **F1-score:** Harmonic mean of precision and recall.
- **Confusion Matrix:** Shows correct and incorrect predictions for benign and malignant classes.

IX. RESULTS AND DISCUSSION

The trained EfficientNet-B0 model was evaluated on the independent test set of **2,314 images**.

Metric	Result
Accuracy	**86.47%**
Malignant Recall	**84.92%**
Malignant F1-score	**70.05%**

The confusion matrix showed **1,635 benign** and **366 malignant** images correctly classified. **248 benign** images were classified as malignant, while **65 malignant** images were classified as benign. The 65 false-negative cases indicate that further improvement is required before clinical use.

Note: Grad-CAM and LIME were not yet applied to the reported experimental results, so didn't **not** claim XAI results in this section yet.

X. APPLICATIONS

The proposed system can be used for:

Multiple roles are served by the proposed system:

- **Image-based screening** of cutaneous lesions.
- **Clinical decision making support** providing predictions and accompanying confidence metrics.
- **Explainable diagnostic support** using Grad-CAM and LIME.
- **Web-based** availability allowing users to submit images and view results.
- **Teledermatology assistance** in areas with scarce resources for dermatologists.

The system is meant as a **supporting tool, not as a replacement for professional diagnostics**.

X. CONCLUSION

EfficientNet-B0 was used for the initial classification into benign and malignant skin lesions in the proposed system. In the testing on an independent test set, the system demonstrated **an accuracy of 86.47%, malignant recall of 84.92% and malignant F1 score of 70.05%.** This proves that there is considerable promise in using deep learning for the initial dermatological diagnostics, but the false negative rate necessitates additional development.

The addition of **Grad-CAM, LIME, predictions with confidence and web-based decision support** is planned.

XII. FUTURE WORK

- Integrate **Grad-CAM and LIME** for visual explanation of model predictions.
- Add **confidence scores** and preliminary decision-support information.
- Complete the **web-based deployment** of the trained model.
- Evaluate model performance across **different skin-tone groups** when suitable data are available.
- Improve the model using larger and more diverse datasets.
- Conduct further validation before considering any real-world clinical use.

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